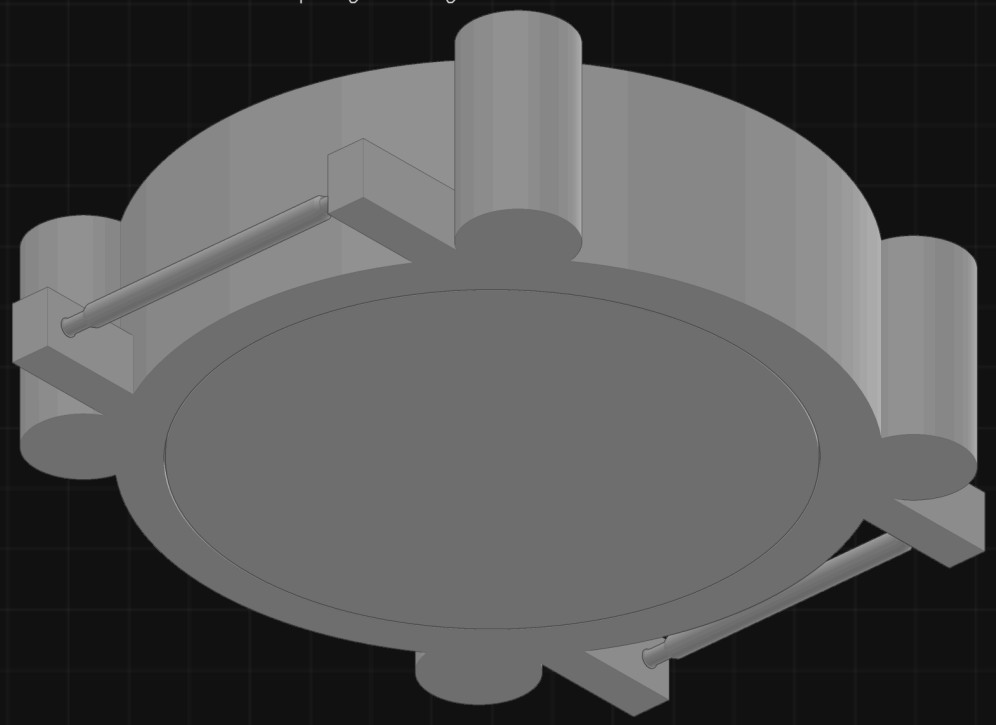
Rev A | 2026-10-24 | UHD-WATCH-DS

DATASHEET · REFERENCE SYSTEM

Wearable Watch

Reference smartwatch: Raytac MDBT50Q-1MV2 (nRF52840, BLE), 1.28-inch round 240 x 240 display (ER-TFT1.28-1, GC9A01), BMI270 IMU, MAX30102 heart rate, DRV2605L with a VG0832012 LRA, MAX17048 fuel gauge, MCP73831 charger from USB-C, 200 mAh LiPo (LP402030), two side buttons, printed PETG case with 22 mm spring-bar lugs.



Assembled from UHD mates (10 placed bodies, 8 fasteners); parts held inside placed by the project's placement hints.

DISPLAY	ACTIVE AREA	BATTERY
240–240	**32.4** mm	**200** mAh
1.28-inch round IPS	display diameter	1S LiPo

PEAK CURRENT	CHARGE CURRENT	ALL-UP WEIGHT
0.177 A	**0.092** A	**12.2** g[A]
with the firmware limits	USB-C, max	parts with a stated mass

ABOUT THIS DOCUMENT

Every value is queried from the UHD model at build time and re-checked by `uhd-tech-docs-verify`. Markers show where a value is not a cited source value.

[A] Assumption: the model marks this value as an assumption (assumption trait)

[D] Derived: computed from model values; formula in the provenance appendix

[–] Not in model: the model does not state this value (data gap)

Unmarked values are cited from the part's sources (sources.json) or are design values of the model.

00 The product

WHAT IT IS

A small, open, fully described smartwatch: a round colour display, motion and heart-rate sensing, haptic alerts and Bluetooth LE to a phone.

FEATURES

- A 1.28-inch round 240×240 colour display
- Step counting, wrist-raise wake and heart rate on demand
- Haptic alerts, and two buttons on the case side
- Bluetooth LE to a phone
- A 200 mAh LiPo cell charged from USB-C; target of 5 days or more of typical use
- A printed case for a standard 22 mm two-piece strap; splash resistant, no ingress rating claimed

WHAT IT DOES

ID	FUNCTION	DONE BY
F1	Show time, notifications and sensor readings on a 1.28-inch round 240×240 colour display	ER-TFT1.28-1 (GC9A01), SPI
F2	Count steps and detect wrist raise to wake the display	BMI270 accelerometer features
F3	Measure heart rate (and SpO2 as a raw signal, not a medical value) on demand and periodically	MAX30102 through a window in the back cover
F4	Vibrate for alerts and confirmations	DRV2605L + 8 mm LRA
F5	Exchange data with a phone over Bluetooth LE	nRF52840 module
F6	Report battery state of charge	MAX17048
F7	Charge from USB-C, and update firmware over USB	MCP73831, USB-C receptacle, nRF52840 USB
F8	Two buttons on the case side: wake/select and back	2 × side-push tact switches

REQUIREMENTS

REQUIREMENT	HOW IT IS SHOWN
Battery: One LiPo cell, about 200 mAh	*Not stated*
Battery life, typical use: ≥ 5 days	*Not stated*
Battery life, standby (BLE connected, screen off): ≥ 30 days	*Not stated*
Charge time: ≤ 3.5 h from empty	*Not stated*
Charging: USB-C 5 V, no USB Power Delivery	*Not stated*
Case envelope: Round case body ≤ 46 mm diameter (lugs excluded), ≤ 13.5 mm thick, 22 mm between lugs	*Not stated*
Weight: ≤ 40 g without strap	*Not stated*
Ingress: Splash resistant (rain, hand washing). No IP rating is claimed and none is tested	*Not stated*
Strap: Standard 22 mm two-piece strap on 1.5 mm spring bars	*Not stated*
Operating temperature: 0 °C to 40 °C	*Not stated*

From docs/brief.md. 10 of 10 do not say how they are shown.

01 Overview

Reference smartwatch: Raytac MDBT50Q-1MV2 (nRF52840, BLE), 1.28-inch round 240 x 240 display (ER-TFT1.28-1, GC9A01), BMI270 IMU, MAX30102 heart rate, DRV2605L with a VG0832012 LRA, MAX17048 fuel gauge, MCP73831 charger from USB-C, 200 mAh LiPo (LP402030), two side buttons, printed PETG case with 22 mm spring-bar lugs.

ROLE	MODULE	QTY
Case	Watch case	1
Main board	Watch main board	1
LiPo cell	LiPol LP402030 200 mAh 3.7 V LiPo with PCM	1
Haptic LRA	Vybronics VG0832012 8 mm Z-axis coin LRA	1
Spring bar (12 o'clock)s	Esslinger 82.2225 spring bar, 22 mm, ø1.50 mm, double flange	2
Display lens	Display lens, 38 × 0.5 mm PMMA	1
Heart-rate window cover	Heart-rate window cover, 4.0 × 8.2 × 0.5 mm PMMA	1
Display adhesive ring	Display adhesive ring, 0.25 mm	1
Cell foam pad	Cell foam pad, 0.25 mm double-sided	1
LRA adhesive pad	LRA adhesive pad, 0.25 mm, ø8 mm	1
Window-cover adhesive	Clear UV-curing adhesive for the window covers	1

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SYSTEM ARCHITECTURE

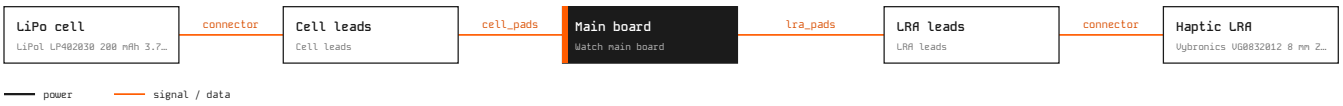


Figure 1 Power and signal links as stored in the model (validated with validateLinks); labels are the hub-side interfaces.

WATCH MAIN BOARD ARCHITECTURE

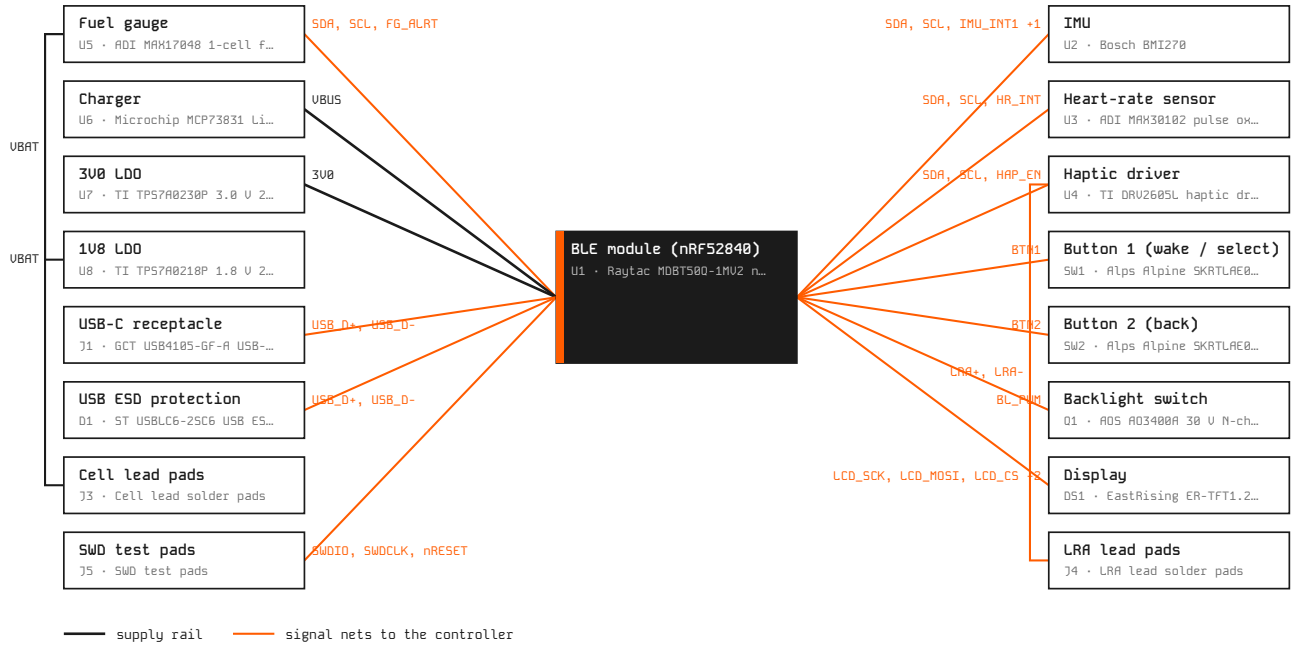


Figure 2 The board's components (passives left out) and the nets that join them to its controller, from the board module's net links. Parts with no net to the controller hang off the part they share a rail with.

02 Specifications

PARAMETER	VALUE
DISPLAY	
Panel	EastRising ER-TFT1.28-1 1.28" round TFT
Technology	IPS TFT LCD, white LED backlight
Resolution	240–240
Active area	32.4 mm
Brightness typical	400 cd/m²
Controller	GC9A01 (GC9A01A)
SENSORS AND HAPTICS	
IMU	Bosch BMI270
Accelerometer ranges	2 × 4 × 8 × 16 g
Gyroscope ranges	125 × 250 × 500 × 1000 × 2000 °/s
Heart rate	ADI MAX30102 pulse oximeter / heart-rate module (OESIP-14)
Heart-rate ADC	18 bit
Haptic driver	TI DRV2605L haptic driver (VSSOP-10)
Actuator	Vybronics VG0832012 8 mm Z-axis coin LRA

PARAMETER	VALUE
POWER	
Cell	LiPol LP402030 200 mAh 3.7 V LiPo with PCM
Capacity	200 mAh
Cell voltage	3.70 V
Continuous discharge	0.2 A
Cell mass	4 g
Charger	Microchip MCP73831 Li-ion charger, 4.20 V (SOT-23-5)
Charge current (max)	0.0917 A
Fuel gauge	ADI MAX17048 1-cell fuel gauge (TDFN-8)

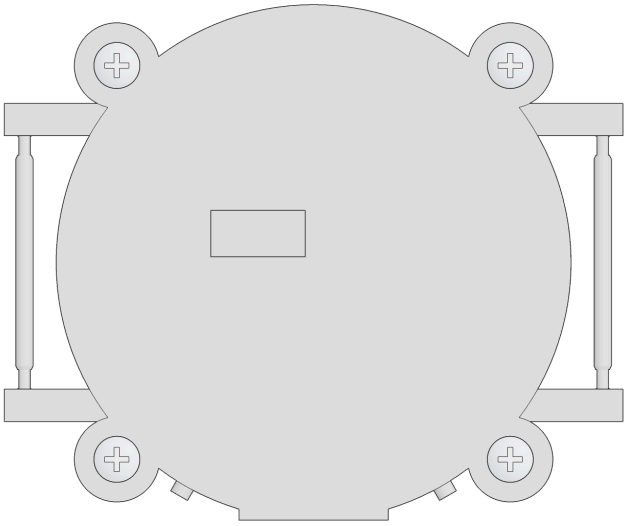
A Assumption: the model marks this value as an assumption (assumption trait)

D Derived: computed from model values; formula in the provenance appendix

– Not in model: the model does not state this value (data gap)

Unmarked values are cited from the part's sources (sources.json) or are design values of the model.

03 Mechanical



ENVELOPE

PARAMETER	VALUE
Case length length across the lugs, max	62 mm
Strap width	22 mm
Case thickness max	13.5 mm
Board diameter	41 mm
Cell thickness	4 mm
Display mass	4 g
All-up weight parts with a stated mass	12.2 g[A]

Figure 3 Top view. (unit: mm)

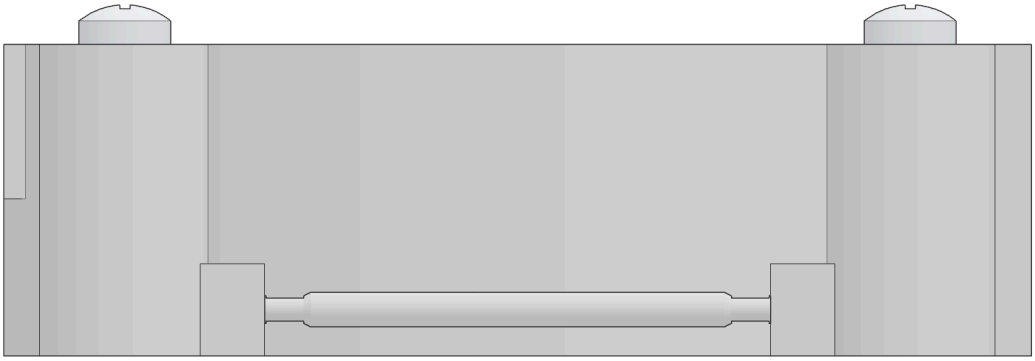


Figure 4 Side view. (unit: mm)

MASS BUDGET

PART	QTY	EACH	BASIS
EastRising ER-TFT1.28-1 1.28" round TFT	1	4 g	stated
LiPol LP402030 200 mAh 3.7 V LiPo with PCM	1	4 g	stated
Vybronics VG0832012 8 mm Z-axis coin LRA	1	1 g	stated
M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless	4	0.19 g	stated
Raytac MDBT50Q-1MV2 nRF52840 module (chip antenna)	1	0.68 g	stated
GCT USB4105-GF-A USB-C receptacle (USB 2.0, top mount)	1	0.68 g	stated
ruthex RX-M2x4 brass heat-set insert, M2 × 4.0 mm	4	0.16 g[A]	CAD × Brass
Esslinger 82.2225 spring bar, 22 mm, ⌀1.50 mm, double flange	2	0.18 g[A]	CAD × Stainless steel
ST USBLC6-2SC6 USB ESD Protection (SOT23-6L)	1	0.017 g	stated
Epson FC-135 32.768 kHz Crystal, 12.5 pF ±20 ppm	1	0.011 g	stated

All-up weight 12.2 g[A] (systemMass), of which 1.0 g[A] is computed from assumed material or nominal CAD geometry. Not included, no mass in the model: Watch case ×1, Bosch BMI270 ×1, ADI MAX30102 pulse oximeter / heart-rate module (OESIP-14) ×1, TI DRV2605L haptic driver (VSSOP-10) ×1, ADI MAX17048 1-cell fuel gauge (TDFN-8) ×1, Microchip MCP73831 Li-ion charger, 4.20 V (SOT-23-5) ×1, TI TPS7A0230P 3.0 V 200 mA LDO (SOT-23-5) ×1, TI TPS7A0218P 1.8 V 200 mA LDO (SOT-23-5) ×1, Alps Alpine SKRTLAE010 side-push TACT switch ×2, AOS AO3400A 30 V N-channel MOSFET (SOT23) ×1, Cell lead solder pads (2) ×1, LRA lead solder pads (2) ×1, SWD test pads (5) ×1, Murata GRM188 MLCC (1608M / 0603, X5R / X7R) ×5, Murata GRM155 MLCC (1005M / 0402, X5R / X7R) ×13, Murata GRM1555C1H200JA01D 20 pF C0G 50 V 0402 ×2, Yageo RC0402 Chip Resistor (RC_L, 0402) ×11, Cell leads ×1, LRA leads ×1, Display lens, 38 × 0.5 mm PMMA ×1, Heart-rate window cover, 4.0 × 8.2 × 0.5 mm PMMA ×1, Display adhesive ring, 0.25 mm ×1, Cell foam pad, 0.25 mm double-sided ×1, LRA adhesive pad, 0.25 mm, ⌀8 mm ×1, Clear UV-curing adhesive for the window covers ×1. Rows in order of total mass (quantity × each).

INTERFACE GEOMETRY

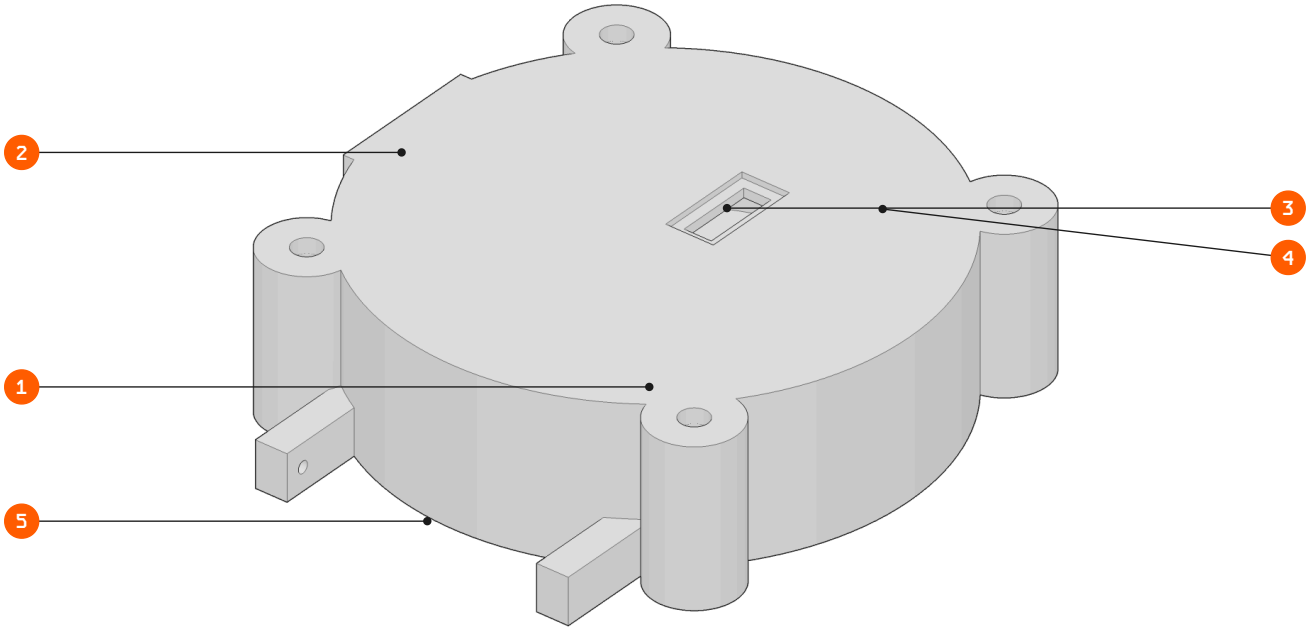


Figure 5 Structural interfaces of the frame and top plate (accent), from each interface's geometry frame and CAD feature binding.

#	INTERFACE	PATTERN	FASTENER	MATES WITH	BINDING
1	**Display window and lens** case.display_window	—⁻ × on —⁻	—⁻	Display lens, 38 × 0.5 mm PMMA	D1
2	**USB-C opening** case.usb_opening	—⁻ × on —⁻	—⁻		D1
3	**Heart-rate window** case.hr_window	—⁻ × on —⁻	—⁻	Heart-rate window cover, 4.0 × 8.2 × 0.5 mm PMMA	D1
4	**Lugs at 12 o'clock** case.lugs_12	—⁻ × on —⁻	—⁻	Esslinger 82.2225 spring bar, 22 mm, ø1.50 mm, double flange	D1
5	**Lugs at 6 o'clock** case.lugs_6	—⁻ × on —⁻	—⁻	Esslinger 82.2225 spring bar, 22 mm, ø1.50 mm, double flange	D1

Binding: D1 named CAD feature, D2 per-interface artifact, D3 procedural from parameters (docs/geometry-artifacts.md).

04 Interfaces and pinout

Allocation of the Watch main board (`board`) interfaces, pad to pad, from the stored links. Crossovers (TX→RX) are marked ×. The link state is the result of pairwise DRC.

HUB INTERFACE	PADS (HUB → PEER)	PEER	PROTOCOL	PARAMETERS	STATE
Cell leads on J3	1 → P1 2 → P2	Cell leads Soldered end (J3)	CONNECTOR	—	CONFIGURED
LRA leads on J4	1 → P1 2 → P2	LRA leads Soldered end (J4)	CONNECTOR	—	CONFIGURED

Pin allocation of the Watch main board's controller, from the board module's nets: every controller pin on a net other than ground, with what else is on that net. Passives are listed by their role.

PAD	PIN	NET	ALSO ON THE NET
17	P0.00/XL1	**XL1**	Y1, C4
18	P0.01/XL2	**XL2**	Y1, C5
22	P0.06	**FG_ALRT**	U5 ALRT (Fuel gauge) R7
24	P0.08	**CHG_STAT**	R10, R11
28	VDD	**3V0**	U7 OUT (3V0 LDO) U2 VDD (IMU) U2 VDDIO (IMU) U2 CSB (IMU) DS1 VDD (Display) J5 VTREF (3V0) (SWD test pads) R4, R5, R6, R7, C1, C2, C7, C17, C18, C20
30	VDDH	**3V0**	U7 OUT (3V0 LDO) U2 VDD (IMU) U2 VDDIO (IMU) U2 CSB (IMU) DS1 VDD (Display) J5 VTREF (3V0) (SWD test pads) R4, R5, R6, R7, C1, C2, C7, C17, C18, C20
32	VBUS	**VBUS**	J1 VBUS (USB-C receptacle) D1 VBUS (USB ESD protection) U6 VDD (Charger) C3, C15, C19
34	D-	**USB_D-**	J1 Dn1 (USB-C receptacle) J1 Dn2 (USB-C receptacle) D1 I/O2 (USB ESD protection)
35	D+	**USB_D+**	J1 Dp1 (USB-C receptacle) J1 Dp2 (USB-C receptacle) D1 I/O1 (USB ESD protection)
36	P0.14	**LCD_MOSI**	DS1 SDA (Display)
37	P0.13	**LCD_SCK**	DS1 SCL (Display)
38	P0.16	**LCD_DC**	DS1 D/C (Display)
39	P0.15	**LCD_CS**	DS1 CS (Display)
40	P0.18/nRESET	**nRESET**	J5 nRESET (SWD test pads)
41	P0.17	**LCD_RST**	DS1 RESET (Display)
42	P0.19	**HR_INT**	U3 INT (Heart-rate sensor) R6
43	P0.21	**HAP_EN**	U4 EN (Haptic driver)
44	P0.20	**BL_PWM**	Q1 G (Backlight switch) R8
45	P0.23	**IMU_INT2**	U2 INT2 (IMU)
46	P0.22	**IMU_INT1**	U2 INT1 (IMU)

CONTROLLER PINOUT (CONTINUED)

PAD	PIN	NET	ALSO ON THE NET
48	P0.24	**SDA**	U2 SDx (IMU) U3 SDA (Heart-rate sensor) U4 SDA (Haptic driver) U5 SDA (Fuel gauge) R4
49	P0.25	**SCL**	U2 SCx (IMU) U3 SCL (Heart-rate sensor) U4 SCL (Haptic driver) U5 SCL (Fuel gauge) R5
50	P1.02	**BTN1**	SW1 1 (Button 1 (wake / select)) SW1 3 (Button 1 (wake / select))
51	SWDIO	**SWDIO**	J5 SWDIO (SWD test pads)
53	SWDCLK	**SWDCLK**	J5 SWDCLK (SWD test pads)
56	P1.04	**BTN2**	SW2 1 (Button 2 (back)) SW2 3 (Button 2 (back))

05 Electrical characteristics



Figure 6 checkSystem supply_budget.

POWER RAILS

RAIL	VOLTAGE	CAPACITY	KNOWN LOAD	LOADS	UNBUDGETED
OUT+ board:lra_pos	0.1–1.9 V	0.008 W	0 W	lra:lead_pos	lra:lead_pos
OUT- board:lra_neg	0.1–1.9 V	0.008 W	0 W	lra:lead_neg	lra:lead_neg
3V0 · Regulated 3.0 V output (OUT + GND) board/u7:vout	3.0 V	0.6 W	0 W	board/u1:pin_28, board/u1:pin_30, board/u2:pin_8, board/u2:pin_5, board/ds1:pin_4	board/u1:pin_28, board/u1:pin_30, board/u2:pin_8, board/u2:pin_5, board/ds1:pin_4
VBAT · Battery charge output (VBAT + VSS) board/u6:battery_charge	4.2 V[A]	1.7 W	0.604 W	board/u3:led_supply, board/u4:supply, board/u5:supply, board/u7:vin, board/u8:vin, board/j3:battery_in	board/u3:led_supply, board/u7:vin, board/u8:vin
1V8 · Regulated 1.8 V output (OUT + GND) board/u8:vout	1.8 V	0.36 W	0.0022 W	board/u3:core_supply	—
VBAT · Charge out (+ / -): MCP73831, 4.20 V, 83 mA board/j3:charge_out	4.2 V	0.312 W	0.0018 W	board/u3:led_supply, board/u4:supply, board/u5:supply, board/u7:vin, board/u8:vin	board/u3:led_supply, board/u7:vin, board/u8:vin

From checkSystem rule supply_budget: capacity = rail voltage × current rating; loads that state no draw cannot be budgeted. A branch rail (supplied_from) is budgeted on the output that feeds it; A marks a relation assumed, not cited.

SUPPLY INPUTS

MODULE	INPUT	VOLTAGE RANGE	MIN. SUPPLY POWER
Raytac MDBT50Q-1MV2 nRF52840 module (chip antenna)	VDD 28	3.0 V[A]	—
Raytac MDBT50Q-1MV2 nRF52840 module (chip antenna)	VDDH 30	1.7–5.5 V[A]	—
Raytac MDBT50Q-1MV2 nRF52840 module (chip antenna)	VBUS 32	5.0 V[A]	—
Bosch BMI270	VDDIO 5	1.8 V	—
Bosch BMI270	VDD 8	1.8 V	—
ADI MAX30102 pulse oximeter / heart-rate module (OESIP-14)	VLED+ 9	3.3 V[A]	—
ADI MAX30102 pulse oximeter / heart-rate module (OESIP-14)	VLED+ 10	3.3 V[A]	—
ADI MAX30102 pulse oximeter / heart-rate module (OESIP-14)	VDD 11	1.8 V[A]	—
ADI MAX30102 pulse oximeter / heart-rate module (OESIP-14)	Core supply (VDD + GND)	1.8 V[A]	—
ADI MAX30102 pulse oximeter / heart-rate module (OESIP-14)	LED supply (VLED+ + PGND)	3.3 V[A]	—
TI DRV2605L haptic driver (VSSOP-10)	VDD/NC 6	2.0–5.2 V	—
TI DRV2605L haptic driver (VSSOP-10)	VDD 10	2.0–5.2 V	—
TI DRV2605L haptic driver (VSSOP-10)	Supply (VDD + GND)	2.0–5.2 V	—
ADI MAX17048 1-cell fuel gauge (TDFN-8)	VDD 3	2.5–4.5 V[A]	—
ADI MAX17048 1-cell fuel gauge (TDFN-8)	Cell supply and sense (VDD + GND + EP)	2.5–4.5 V[A]	—
Microchip MCP73831 Li-ion charger, 4.20 V (SOT-23-5)	VDD 4	5.0 V[A]	—
Microchip MCP73831 Li-ion charger, 4.20 V (SOT-23-5)	Charger input (VDD + VSS)	5.0 V[A]	—
TI TPS7A0230P 3.0 V 200 mA LDO (SOT-23-5)	IN 1	1.5–6.0 V	—
TI TPS7A0230P 3.0 V 200 mA LDO (SOT-23-5)	Input supply (IN + GND)	1.5–6.0 V	—
TI TPS7A0218P 1.8 V 200 mA LDO (SOT-23-5)	IN 1	1.5–6.0 V	—
TI TPS7A0218P 1.8 V 200 mA LDO (SOT-23-5)	Input supply (IN + GND)	1.5–6.0 V	—
ST USBLC6-2SC6 USB ESD Protection (SOT23-6L)	VBUS 5	5.0 V[A]	—
ST USBLC6-2SC6 USB ESD Protection (SOT23-6L)	VBUS clamp (VBUS + GND)	5.0 V[A]	—
EastRising ER-TFT1.28-1 1.28" round TFT	VDD 4	2.8 V	—
Cell lead solder pads (2)	VBAT (+) 1	3.7 V	—
Cell lead solder pads (2)	Battery in (+ / -): the watch's loads	3.7 V	—
LiPol LP402030 200 mAh 3.7 V LiPo with PCM	Lead + (red, AWG30 UL1007) +	3.7 V	—
LiPol LP402030 200 mAh 3.7 V LiPo with PCM	Charge input (+ / -), 4.2 V CC/CV, 100 mA max	4.2 V	—
Vybronics VG0832012 8 mm Z-axis coin LRA	Lead + (red) +	0.1–1.9 V	—
Vybronics VG0832012 8 mm Z-axis coin LRA	Lead - (blue) -	0.1–1.9 V	—
Vybronics VG0832012 8 mm Z-axis coin LRA	LRA drive (+ / -), 1.8 Vrms 235 Hz	0.1–1.9 V	—

Stored links: 12; configured: 12. Diagnostics on the nominal build: info supply_budget, info supply_budget, info fastener_torque.

06 Performance

TEST DATA

Charge used per typical day

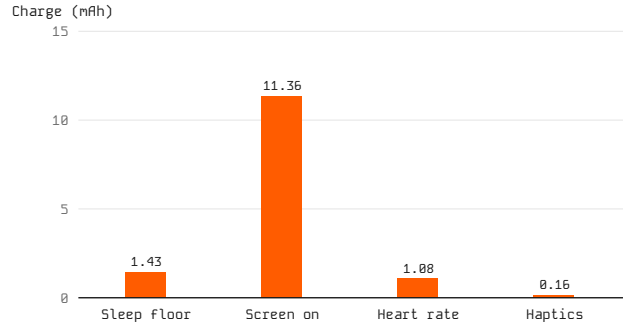


Figure 7 From the power budget's use profile: the screen dominates a typical day.

Peak current stack-up

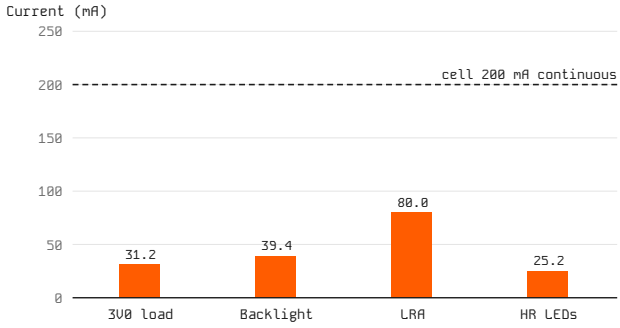


Figure 8 Instantaneous currents from VBAT with the firmware limits; their sum is the peak.

BATTERY AND POWER (ESTIMATES)

FIGURE	VALUE	BASIS
Battery life, typical use	12.1 days	35 min screen, HR every 10 min, 20 alerts a day
Battery life, standby	119 days	BLE connected, screen off
Charge per typical day	14.03 mAh/day	
Sleep floor	59.5 µA	
Peak current	177 mA	instantaneous, with the firmware limits
Charge current, max	91.7 mA	
Charge time	2.4 h	estimate, from empty

Computed by src/power-budget.ts from datasheet currents and an assumed use profile (docs/power-budget.md). Not measured: a discharge test replaces these.

07 Design checks

CHECKSYSTEM

The nominal build and each fault scenario in `scenarios.ts` run through `checkSystem` (the system and each circuit board in it). A fault scenario is expected to raise at least one error or warning.

SCENARIO	DIAGNOSTICS	FINDINGS (ERRORS AND WARNINGS)
Nominal build nominal	0 ERR 0 WARN 7 INFO	no errors or warnings
MAX30102 on the 3.0 V rail hr-sensor-on-3v0	2 ERR 0 WARN 6 INFO	LINK_STATE board: Core supply (VDD + GND) ↔ Regulated 3.0 V output (OUT + GND): incompatible — voltage: 1.7–2 V vs 3 V (ranges do not overlap) NET board: u3:pin_11 (VDD) is rated 1.7–2 V but net v3v0 (3V0) is 3 V.
Display VDD on the cell display-on-vbat	3 ERR 0 WARN 7 INFO	LINK_STATE board: VBAT ↔ VDD: incompatible — voltage: 3.4–4.2 V vs 2.5–3.3 V (ranges do not overlap) LINK_STATE board: VDD ↔ VBAT (+): incompatible — voltage: 2.5–3.3 V vs 3.4–4.2 V (ranges do not overlap) NET board: ds1:pin_4 (VDD) is rated 2.5–3.3 V but net vbat (VBAT) is 3.4–4.2 V.
I2C bus without pull-ups no-i2c-pullups	0 ERR 2 WARN 7 INFO	BUS_PULLUP board: net sda carries I2C (u2:pin_14) but no resistor pulls it up to a supply. BUS_PULLUP board: net scl carries I2C (u2:pin_13) but no resistor pulls it up to a supply.
Cell leads soldered the wrong way round cell-leads-reversed	1 ERR 0 WARN 7 INFO	LINK_STATE Battery output (+ / -), 3.7 V 200 mAh ↔ Battery in (+ / -): the watch's loads: incompatible — lead_pos is wired to pin_2, but power pairs it with pin_1; lead_neg is wired to pin_1, but power pairs it with pin_2
3V0 LDO output tied to VBAT ldo-output-to-vbat	5 ERR 0 WARN 7 INFO	LINK_STATE board: LED supply (VLED+ + PGND) ↔ Regulated 3.0 V output (OUT + GND): incompatible — voltage: 3.1–5 V vs 2.955–3.045 V (ranges do not overlap) LINK_STATE board: Regulated 3.0 V output (OUT + GND) ↔ Battery in (+ / -): the watch's loads: incompatible — power gnd: u7:vout pin_2 and j3:battery_in pin_2 are not wired to each other NET board: net vbat (VBAT) is driven by 2 supplies (u6:pin_3, u7:pin_5); outputs in parallel fight each other.
Charger STAT straight to the MCU stat-overdrive	1 ERR 1 WARN 7 INFO	NET board: u6:pin_1 (STAT) drives net chg_stat (CHG_STAT) up to its supply VBUS (5 V), but u1:pin_24 (P0.08) is rated to 3.6 V: divide, clamp or use an open-drain output. NET board: net stat (STAT) joins no pins.

08 Bill of materials

DOCS/WEARABLE-WATCH-BOM.JSON

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
1	ADI MAX17048 1-cell fuel gauge (TDFN-8) adi-max17048g-t10	Analog Devices (Maxim Integrated) MAX17048G+T10	1	PURCHASE	—	— A 2 · gaps 3
2	ADI MAX30102 pulse oximeter / heart-rate module (OESIP-14) adi-max30102efd	Analog Devices (Maxim Integrated) MAX30102EFD+T	1	PURCHASE	—	— A 2 · gaps 3
3	Alps Alpine SKRTLAE010 side-push TACT switch alps-skrtlae010	Alps Alpine SKRTLAE010	2	PURCHASE	—	not audited A 3 · gaps 3
4	AOS AO3400A 30 V N-channel MOSFET (SOT23) aos-ao3400a	Alpha & Omega Semiconductor AO3400A	1	PURCHASE	—	audited 20/20 A 2 · gaps 5
5	Bosch BMI270 bosch-bmi270	Bosch Sensortec BMI270	1	PURCHASE	—	audited 19/19 A 1 · gaps 2
6	Cell foam pad, 0.25 mm double-sided watch-cell-foam-pad	—	1	PURCHASE	—	— A 1 · gaps 1
7	Clear UV-curing adhesive for the window covers watch-cover-adhesive	—	1	PURCHASE	—	— A 1 · gaps 1
8	Display adhesive ring, 0.25 mm watch-display-adhesive-ring	—	1	PURCHASE	—	— A 1 · gaps 1
9	EastRising ER-TFT1.28-1 1.28" round TFT eastrising-er-tft128-1	EastRising Technology (BuyDisplay) ER-TFT1.28-1	1	PURCHASE	4 g	audited 19/19 A 6 · gaps 6
10	Epson FC-135 32.768 kHz Crystal, 12.5 pF ±20 ppm epson-fc-135-32k768-12p5	Seiko Epson Q13FC13500004	1	PURCHASE	0.011 g	audited 32/35 A 1 · gaps 3
11	Esslinger 82.2225 spring bar, 22 mm, ø1.50 mm, double flange esslinger-82-2225-spring-bar	Esslinger 82.2225	2	PURCHASE	0.18 g[A]	audited 9/9 A 2 · gaps 4
12	GCT USB4105-GF-A USB-C receptacle (USB 2.0, top mount) gct-usb4105-gf-a	GCT USB4105-GF-A	1	PURCHASE	0.68 g	audited 19/19 A 2 · gaps 4
13	LiPol LP402030 200 mAh 3.7 V LiPo with PCM lipol-lp402030	LiPol Battery Co., Ltd. LP402030	1	PURCHASE	4 g	audited 14/14 A 3 · gaps 6
14	LRA adhesive pad, 0.25 mm, ø8 mm watch-lra-adhesive-pad	—	1	PURCHASE	—	— A 1 · gaps 1
15	M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless iso-7045-m2x6-a2	Generic (ISO 7045 / DIN 7985 Z) SPP-M2-6-A2	4	PURCHASE	0.19 g	not audited A 2 · gaps 4
16	Microchip MCP73831 Li-ion charger, 4.20 V (SOT-23-5) microchip-mcp73831t-2aci-ot	Microchip Technology MCP73831T-2ACI/OT	1	PURCHASE	—	audited A 2 · gaps 3
17	Murata GRM155 MLCC (1005M / 0402, X5R / X7R) 4.7 µF ±20 % murata-grm155	Murata GRM155R60J475ME47D	1	PURCHASE	—	audited 10/10 A 2 · gaps 3
18	Murata GRM155 MLCC (1005M / 0402, X5R / X7R) 1 µF ±10 % murata-grm155	Murata GRM155R61A105KE15D	6	PURCHASE	—	audited 10/10 A 2 · gaps 3
19	Murata GRM155 MLCC (1005M / 0402, X5R / X7R) 100 nF ±10 % murata-grm155	Murata GRM155R71C104KA88D	6	PURCHASE	—	audited 10/10 A 2 · gaps 3

Rows as written by the document build (docs/wearable-watch-bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

BILL OF MATERIALS (CONTINUED)

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
20	Murata GRM1555C1H200JA01D 20 pF C0G 50 V 0402 20 pF ±5 % murata-grm1555c1h200ja01	Murata GRM1555C1H200JA01D	2	PURCHASE	—	audited 8/8 A 2 · gaps 2
21	Murata GRM188 MLCC (1608M / 0603, X5R / X7R) 10 µF ±10 % murata-grm188	Murata GRM188R61A106KE69D	1	PURCHASE	—	audited 8/8 A 2 · gaps 3
22	Murata GRM188 MLCC (1608M / 0603, X5R / X7R) 4.7 µF ±10 % murata-grm188	Murata GRM188R61A475KE15D	4	PURCHASE	—	audited 8/8 A 2 · gaps 3
23	Raytac MDBT50Q-1MV2 nRF52840 module (chip antenna) raytac-mdbt50q-1mv2	Raytac Corporation MDBT50Q-1MV2	1	PURCHASE	0.68 g	audited 20/20 A 3 · gaps 3
24	ruthex RX-M2x4 brass heat-set insert, M2 x 4.0 mm ruthex-rx-m2x4	ruthex GE-M2x04-001	4	PURCHASE	0.16 g[A]	audited 21/21 A 1 · gaps 3
25	ST USBLC6-2SC6 USB ESD Protection (SOT23-6L) st-usblc6-2sc6	STMicroelectronics USBLC6-2SC6	1	PURCHASE	0.017 g	not audited A 2 · gaps 5
26	TI DRV2605L haptic driver (VSSOP-10) ti-drv2605ldgs	Texas Instruments DRV2605LDGSR	1	PURCHASE	—	— A 1 · gaps 3
27	TI TPS7A0218P 1.8 V 200 mA LDO (SOT-23-5) ti-tps7a0218pdbv	Texas Instruments TPS7A0218PDBVR	1	PURCHASE	—	audited 18/18 A 1 · gaps 4
28	TI TPS7A0230P 3.0 V 200 mA LDO (SOT-23-5) ti-tps7a0230pdbv	Texas Instruments TPS7A0230PDBVR	1	PURCHASE	—	audited 18/18 A 1 · gaps 4
29	Vybronics VG0832012 8 mm Z-axis coin LRA vybronics-vg0832012	Vybronics VG0832012	1	PURCHASE	1 g	— A 2 · gaps 2
30	Watch case watch-case	—	1	PURCHASE	—	— A 0 · gaps 0
31	Yageo RC0402 Chip Resistor (RC_L, 0402) 100 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-07100KL	1	PURCHASE	—	audited 9/9 A 2 · gaps 2
32	Yageo RC0402 Chip Resistor (RC_L, 0402) 10 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-0710KL	2	PURCHASE	—	audited 9/9 A 2 · gaps 2
33	Yageo RC0402 Chip Resistor (RC_L, 0402) 12 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-0712KL	1	PURCHASE	—	audited 9/9 A 2 · gaps 2
34	Yageo RC0402 Chip Resistor (RC_L, 0402) 22 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-0722KL	2	PURCHASE	—	audited 9/9 A 2 · gaps 2
35	Yageo RC0402 Chip Resistor (RC_L, 0402) 33 ohm ±1 % yageo-rc0402	Yageo RC0402FR-0733RL	1	PURCHASE	—	audited 9/9 A 2 · gaps 2
36	Yageo RC0402 Chip Resistor (RC_L, 0402) 4.7 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-074K7L	2	PURCHASE	—	audited 9/9 A 2 · gaps 2
37	Yageo RC0402 Chip Resistor (RC_L, 0402) 5.1 k ohm ±1 % yageo-rc0402	Yageo RC0402FR-075K1L	2	PURCHASE	—	audited 9/9 A 2 · gaps 2
38	Display lens, 38 × 0.5 mm PMMA watch-display-lens	Laser cut for this project (custom)	1	FABRICATE	—	— A 1 · gaps 1
39	Heart-rate window cover, 4.0 × 8.2 × 0.5 mm PMMA watch-hr-cover	Laser cut for this project (custom)	1	FABRICATE	—	— A 1 · gaps 1

Rows as written by the document build (docs/wearable-watch-bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

BILL OF MATERIALS (CONTINUED)

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
40	Watch main board watch-main-board	—"	1	FABRICATE	—	—" A 2 · gaps 0
41	Cell lead solder pads (2) watch-cell-pads	—"	1	INCL. watch-main-board	—	—" A 0 · gaps 0
42	Cell leads watch-battery-leads	LiPol Battery Co.	1	INCL. lipol-lp402030	—	—" A 1 · gaps 0
43	LRA lead solder pads (2) watch-lra-pads	—"	1	INCL. watch-main-board	—	—" A 0 · gaps 0
44	LRA leads watch-lra-leads	Vybronics	1	INCL. vybronics-vg0832012	—	—" A 1 · gaps 0
45	SWD test pads (5) watch-swd-pads	—"	1	INCL. watch-main-board	—	—" A 0 · gaps 0

Rows as written by the document build (docs/wearable-watch-bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

09 Evidence and assumptions

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
ADI MAX17048 1-cell fuel gauge (TDFN-8)	6	not audited	A QSTRT voltage range: 0-5.5 V A CAD package dimensions: nominal body 2.00 x 2.00 x 0.75 mm, terminals 0.25 x 0.30 mm, exposed pad 1.20 x 0.80 mm w	—
ADI MAX30102 pulse oximeter / heart-rate module (OESIP-14)	4	not audited	A SDA, SCL, INT voltage: 0-6.0 V A CAD cover glass and windows: housing 3.3 x 5.6 x 1.35 mm; cover glass 3.1 x 5.4 x 0.20 mm (0.10 rim inset) on top; wind	—
Alps Alpine SKRTLAE010 side-push TACT switch	5	not audited	A typical wiring: terminal 2 to a GPIO input with a pull-up; terminals 1 and 3 to GND A terminal 1 position: terminal 1 at -X: the left terminal looking at the stem A CAD dimensions not on the drawing: terminal feet 0.15 mm thick, flat on the mounting face, 1.34-2.67 mm behind the frame face	manufacturer CAD; mass; terminal thickness and riser shape
AOS AO3400A 30 V N-channel MOSFET (SOT23)	8	20/20	A pin numbers: pin 1 = G, pin 2 = S, pin 3 = D A CAD package dimensions: nominal values (body D 2.90 x E1 1.60, span E 2.80, lead b 0.40 x c 0.13, foot L 0.45, bod	manufacturer CAD; pin numbers; mass; operating ambient temperature; recommended gate drive range
Bosch BMI270	3	19/19	A CAD pad thickness: 0.02 mm	manufacturer CAD; pad thickness
Cell foam pad, 0.25 mm double-sided	—	not audited	A specification: 0.25 mm double-coated closed-cell foam (PE or PU), acrylic adhesive, die-cut to the cell's	product; supplier
Clear UV-curing adhesive for the window covers	—	not audited	A specification: clear, low-viscosity UV-curing acrylic adhesive for PMMA to PETG, applied at the cover's r	product; supplier
Display adhesive ring, 0.25 mm	—	not audited	A specification: 0.25 mm double-coated tape (thin foam or acrylic carrier), die-cut ring ID 34.0 / OD 35.6	product; supplier
EastRising ER-TFT1.28-1 1.28" round TFT	7	19/19	A mass: 4 g A CAD layer split: backlight 0.80 mm, LCD glass 0.60 mm, polarizer 0.10 mm A CAD tab and FPC neck outline: tab flat end 14.0 mm wide, chamfered to the circle at +-8.6 mm; FPC neck 8.7 mm wide throu A FPC alignment-hole position along the tail: hole centres 4.0 mm from the tail end, 2.0 mm beyond the 2.00 mm window (CAD x = 18.41 mm A CAD tail pad widening: the tail pad is 0.9 mm wider than 11.4 mm on the pin-1 side, from 3.8 mm in from the tail A CAD installed state: Installed as in the wearable watch (display / cell / board, OQ-W5): the FPC leaves the tab	manufacturer CAD; mass; layer stack split; tab and neck outline; SPI clock for the panel (+1)
Epson FC-135 32.768 kHz Crystal, 12.5 pF ±20 ppm	6	32/35	A CAD package dimensions: nominal body 3.2 x 1.5 at the 0.9 mm maximum height; terminal thickness 0.03; ceramic base	manufacturer CAD; package code; thermal resistance

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

EVIDENCE (CONTINUED)

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
Esslinger 82.2225 spring bar, 22 mm, ø1.50 mm, double flange	4	9/9	A end-feature dimensions: tip length 0.8 mm beyond the outer flange; flanges ø1.2 × 0.25 mm with a ø0.9 × 0.25 mm gr A CAD state: free (as supplied): plungers fully extended, overall 25.0 mm	manufacturer CAD; tip length, flange diameter and width, groove, shank diameter, tube wall; mass; stainless grade, spring force, operating temperature
GCT USB4105-GF-A USB-C receptacle (USB 2.0, top mount)	6	19/19	A D+/D- contacts on both sides: the board joins A6 (Dp1) with B6 (Dp2) and A7 (Dn1) with B7 (Dn2) A CAD shell and internals: outer shell a stadium 8.94 × 3.16 mm (0.30 mm wall around the 8.34 × 2.56 opening, bottom	manufacturer CAD; mass (GCT states none; LCSC unit weight used); receptacle internals; lifecycle (GCT product page unreadable)
LiPol LP402030 200 mAh 3.7 V LiPo with PCM	5	14/14	A length includes the PCM: 30 mm overall, of which the last 3.2 mm (along the length) is the folded PCM zone at full A lead side and order: both leads leave the PCM end face next to the '+' side edge, red (+) outermost at y = +9.0 A generated lead geometry: insulation Ø1.1 mm, conductor Ø0.30 mm; installed state (wearable watch): only a 1 mm stub	manufacturer CAD; lead position and order on the end face; lead strip length and conductor size; PCM zone geometry; pulse discharge current, swelling allowance, self-discharge (+1)
LRA adhesive pad, 0.25 mm, ø8 mm	—	not audited	A specification: 0.25 mm double-coated acrylic adhesive (a firm transfer tape, not foam), die-cut ø8 mm to	product; supplier
M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless	5	not audited	A ISO 7045 edition: M2 dimensions from ISO 7045:1994 (second edition) A CAD simplifications: plain Ø2.0 shank (thread major diameter, no thread form) with a flat end; under-head radiu	manufacturer CAD; Pozidriv recess form; tightening torque, proof load, operating temperature; as-rolled end form
Microchip MCP73831 Li-ion charger, 4.20 V (SOT-23-5)	3	audited	A VBAT voltage lower bound: 0 V A CAD package dimensions: midpoints of each drawing range (A2 1.095, A1 0.075, L 0.45, c 0.17, b 0.355); body top 1.	manufacturer CAD; PROG logic-high threshold; VBAT minimum voltage
Murata GRM155 MLCC (1005M / 0402, X5R / X7R)	7	10/10	A terminal designators: pin_1 = the -X termination, pin_2 = the +X termination A CAD package dimensions: nominal 1.0 × 0.5 × 0.5 mm chip; each termination one block 0.25 mm long (middle of e 0.15	manufacturer CAD (none linked from the cited sources; one generated body stands for the series); mass (no unit mass in the cited sources); series lineup (which capacitance exists at which voltage and dielectric is not in the per-part reference sheets; a placement names its exact MPN)

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

EVIDENCE (CONTINUED)

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
Murata GRM1555C1H200JA01D 20 pF C0G 50 V 0402	1	8/8	A terminal designators: pin_1 = the -X termination, pin_2 = the +X termination A CAD package dimensions: nominal 1.0 × 0.5 × 0.5 mm chip; each termination one 0.25 mm block over the full width an	manufacturer CAD; mass
Murata GRM188 MLCC (1608M / 0603, X5R / X7R)	6	8/8	A terminal designators: pin_1 = the -X termination, pin_2 = the +X termination A CAD package dimensions: nominal 1.6 × 0.8 × 0.8 mm chip; each termination one block 0.35 mm long (middle of e 0.2	manufacturer CAD (none linked from the cited sources; one generated body stands for the series); mass (no unit mass in the cited sources); series lineup (which capacitance exists at which voltage and dielectric is not in the per-part reference sheets; a placement names its exact MPN)
Raytac MDBT50Q-1MV2 nRF52840 module (chip antenna)	5	20/20	A GPIO voltage parameter: 1.7-3.6 V on every logic pad A CAD pad size and thickness: 0.6 × 0.4 mm at the §2.2 centres, 0.02 mm thick A CAD shield position: centred in width, 0.4 mm from the pad-edge end	manufacturer CAD (redistributable); module pad size and pad thickness; shield position
ruthex RX-M2x4 brass heat-set insert, M2 × 4.0 mm	10	21/21	A generated body: 12-tooth knurl bands with a 4° flat crest and a V root at a constant twist (upper -50° ove	manufacturer CAD; mass; brass alloy, pull-out force, torque-out, tightening torque, operating temperature, installation temperature
ST USBLC6-2SC6 USB ESD Protection (SOT23-6L)	4	not audited	A VBUS working voltage: 0-5.25 V, nominal 5 V A CAD package dimensions: midpoint of each Table 3 range (body 2.9 × 1.6, span 2.8, lead 0.4 wide × 0.15 thick, foot	manufacturer CAD; datasheet revision; VBUS working voltage; internal I/O pin connection; thermal resistance
TI DRV2605L haptic driver (VSSOP-10)	4	not audited	A CAD package dimensions: nominal = midpoint of each drawing range (body 3.0 × 3.0, span 4.9, lead 0.22 × 0.18, foot	—
TI TPS7A0218P 1.8 V 200 mA LDO (SOT-23-5)	4	18/18	A CAD package dimensions: nominal = midpoint of each drawing range (body 2.9 × 1.6, span 2.8, lead 0.4 wide × 0.15 t	manufacturer CAD; mass; operating ambient temperature; input current draw
TI TPS7A0230P 3.0 V 200 mA LDO (SOT-23-5)	4	18/18	A CAD package dimensions: nominal = midpoint of each drawing range (body 2.9 × 1.6, span 2.8, lead 0.4 wide × 0.15 t	manufacturer CAD; mass; operating ambient temperature; input current draw

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

EVIDENCE (CONTINUED)

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
Vybronics VG0832012 8 mm Z-axis coin LRA	4	not audited	A generated geometry: lead tab 2.4 mm wide, 0.25 mm thick, ending at x = 5.6; UV adhesive drawn at its maximum e A lead side: red '+' at +Y and blue '-' at -Y with the leads along +X, seen from the top	coil inductance; resonant-frequency shift with mounting mass; lead tab width and thickness; lead spacing and insulation diameter
Watch case	—	not audited	—	—
Yageo RC0402 Chip Resistor (RC_L, 0402)	1	9/9	A terminal designators: pin_1 = the -X termination, pin_2 = the +X termination A CAD package dimensions: nominal 1.00 x 0.50 x 0.35 mm chip; each termination one block 0.25 mm long (I2) over the	manufacturer CAD (none linked from the cited sources; one generated body stands for the series); mass (no unit mass in the cited sources)
Display lens, 38 × 0.5 mm PMMA	—	not audited	A material and thickness: clear cast PMMA, 0.5 mm	supplier; sheet grade
Heart-rate window cover, 4.0 × 8.2 × 0.5 mm PMMA	—	not audited	A material and thickness: clear cast PMMA, 0.5 mm	supplier; sheet grade
Watch main board	—	not audited	A Y1 load capacitance: C4 = C5 = 20 pF C0G: 2 x CL (12.5 pF, Y1) - Cpin (4 pF) - Cpcb (1 pF) A pull-up and series values: I2C 4.7 kOhm, HR INT and FG ALRT 10 kOhm, Q1 gate 100 kOhm, backlight 33 Ohm	—
Cell lead solder pads (2)	—	not audited	—	—
Cell leads	—	not audited	A lead insulation diameter: 1.1 mm	—
LRA lead solder pads (2)	—	not audited	—	—
LRA leads	—	not audited	A lead diameter: 0.6 mm	—
SWD test pads (5)	—	not audited	—	—

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

10 Value provenance

This document shows 557 bound values. Each is a query into the UHD model, re-resolved and compared by `uhd-tech-docs-verify`; the full list with sources is in `uhd-watch-ds.values.json` next to the document.

STATUS	VALUES	MEANING
Cited source	73	stated by a source in the part's sources.json
Model design value	403	a design value of a custom module, or a UHD check result
Derived	21	computed from model values; formula in the provenance appendix
Assumption	19	the model marks this value as an assumption (assumption trait)
Not in model	41	the model does not state this value (data gap)

STATUS	VALUE	QUERY	DERIVATION / REASON
A Assumption	12.2 g	derived:mass.all_up	systemMass: Σ quantity × (weight_g, else CAD volume × material density) over every instance; instances with neither are listed, not guessed · excludes 53 part(s) with no mass: Watch case, Bosch BMI270, ADI MAX30102 pulse oximeter / heart-rate module (OESIP-14)
A Assumption	0.16 g	derived:mass.module(ruthex-rx-m2x4)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.02 cm³ × 8.5 g/cm³ (Brass (lead-free, cadmium-free; alloy not stated)); assumption: ruthex names no alloy; 8.5 g/cm³ is within the table's rolled-and
A Assumption	0.18 g	derived:mass.module(esslinger-82-2225-spring-bar)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.02 cm³ × 8 g/cm³ (Stainless steel (grade not stated; 304 / 1.4301 density used)); assumption: Esslinger states only 'stainless steel', so the 304 den
A Assumption	1.0 g	derived:mass.assumed	Σ systemMass entries whose material carries an assumption · inputs: 2
– Not in model	–	inst:case:interfaces[id=display_window].parameters[id=hole_count].value	
– Not in model	–	inst:case:interfaces[id=display_window].parameters[id=hole_spacing].value	
– Not in model	–	inst:case:interfaces[id=display_window].parameters[id=fastener_diameter].value	
– Not in model	–	inst:case:interfaces[id=usb_opening].parameters[id=hole_count].value	
– Not in model	–	inst:case:interfaces[id=usb_opening].parameters[id=hole_spacing].value	
– Not in model	–	inst:case:interfaces[id=usb_opening].parameters[id=fastener_diameter].value	
– Not in model	–	inst:case:interfaces[id=hr_window].parameters[id=hole_count].value	
– Not in model	–	inst:case:interfaces[id=hr_window].parameters[id=hole_spacing].value	
– Not in model	–	inst:case:interfaces[id=hr_window].parameters[id=fastener_diameter].value	
– Not in model	–	inst:case:interfaces[id=lugs_12].parameters[id=hole_count].value	
– Not in model	–	inst:case:interfaces[id=lugs_12].parameters[id=hole_spacing].value	
– Not in model	–	inst:case:interfaces[id=lugs_12].parameters[id=fastener_diameter].value	
– Not in model	–	inst:case:interfaces[id=lugs_6].parameters[id=hole_count].value	

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
– Not in model	—	inst:case:interfaces[id=lugs_6].parameters[id=hole_spacing].value	
– Not in model	—	inst:case:interfaces[id=lugs_6].parameters[id=fastener_diameter].value	
A Assumption	4.2 V	def:microchip-mcp73831t-2aci-ot:interfaces[id=battery_charge].parameters[id=voltage].value	The data sheet gives no minimum VBAT: preconditioning applies below 66.5 % of VREG with no floor, so the pin follows the cell from 0 V up to VREG (4.232 V max). · https://ww1.microchip.com/downloads/aemDocuments/documents/APID/ProductDocuments/DataSheets/MCP73
A Assumption	3.0 V	def:raytac-mdbt50q-1mv2:interfaces[id=pin_28].parameters[id=voltage].value	GPIO high level equals VDD (normal voltage mode, 1.7-3.6 V) or REGOUT0 (high voltage mode, 1.8-3.3 V); the parameter states the union so a 3.0 V or 3.3 V rail checks against it (PS §5.3.1.2). · https://www.raytac.com/download/index.php?index_id=43
A Assumption	1.7–5.5 V	def:raytac-mdbt50q-1mv2:interfaces[id=pin_30].parameters[id=voltage].value	GPIO high level equals VDD (normal voltage mode, 1.7-3.6 V) or REGOUT0 (high voltage mode, 1.8-3.3 V); the parameter states the union so a 3.0 V or 3.3 V rail checks against it (PS §5.3.1.2). · https://www.raytac.com/download/index.php?index_id=43
A Assumption	5.0 V	def:raytac-mdbt50q-1mv2:interfaces[id=pin_32].parameters[id=voltage].value	GPIO high level equals VDD (normal voltage mode, 1.7-3.6 V) or REGOUT0 (high voltage mode, 1.8-3.3 V); the parameter states the union so a 3.0 V or 3.3 V rail checks against it (PS §5.3.1.2). · https://www.raytac.com/download/index.php?index_id=43
A Assumption	3.3 V	def:adi-max30102efd:interfaces[id=pin_9].parameters[id=voltage].value	The datasheet states no recommended operating maximum for these pins, only the 6.0 V absolute maximum for all other pins (p.2), and allows pull-ups to an external voltage (p.28). 6.0 V is used as the ceiling for net checks; keep pull-ups well below it (the app
A Assumption	3.3 V	def:adi-max30102efd:interfaces[id=pin_10].parameters[id=voltage].value	The datasheet states no recommended operating maximum for these pins, only the 6.0 V absolute maximum for all other pins (p.2), and allows pull-ups to an external voltage (p.28). 6.0 V is used as the ceiling for net checks; keep pull-ups well below it (the app
A Assumption	1.8 V	def:adi-max30102efd:interfaces[id=pin_11].parameters[id=voltage].value	The datasheet states no recommended operating maximum for these pins, only the 6.0 V absolute maximum for all other pins (p.2), and allows pull-ups to an external voltage (p.28). 6.0 V is used as the ceiling for net checks; keep pull-ups well below it (the app
A Assumption	1.8 V	def:adi-max30102efd:interfaces[id=core_supply].parameters[id=voltage].value	The datasheet states no recommended operating maximum for these pins, only the 6.0 V absolute maximum for all other pins (p.2), and allows pull-ups to an external voltage (p.28). 6.0 V is used as the ceiling for net checks; keep pull-ups well below it (the app
A Assumption	3.3 V	def:adi-max30102efd:interfaces[id=led_supply].parameters[id=voltage].value	The datasheet states no recommended operating maximum for these pins, only the 6.0 V absolute maximum for all other pins (p.2), and allows pull-ups to an external voltage (p.28). 6.0 V is used as the ceiling for net checks; keep pull-ups well below it (the app
A Assumption	2.5–4.5 V	def:adi-max17048g-t10:interfaces[id=pin_3].parameters[id=voltage].value	The electrical characteristics give the -0.3 to +5.5 V data I/O range for SCL, SDA and ALRT only; QSTRT has just the -0.3 to +6 V absolute maximum. The data I/O range is used as the conservative operating limit. · https://www.analog.com/media/en/technical-docu

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
A Assumption	2.5–4.5 V	def:adi-max17048g-t10:interfaces[id=supply].parameters[id=voltage].value	The electrical characteristics give the -0.3 to +5.5 V data I/O range for SCL, SDA and ALRT only; QSTRT has just the -0.3 to +6 V absolute maximum. The data I/O range is used as the conservative operating limit. · https://www.analog.com/media/en/technical-docu
A Assumption	5.0 V	def:microchip-mcp73831t-2aci-ot:interfaces[id=pin_4].parameters[id=voltage].value	The data sheet gives no minimum VBAT: preconditioning applies below 66.5 % of VREG with no floor, so the pin follows the cell from 0 V up to VREG (4.232 V max). · https://ww1.microchip.com/downloads/aemDocuments/documents/APID/ProductDocuments/DataSheets/MCP73
A Assumption	5.0 V	def:microchip-mcp73831t-2aci-ot:interfaces[id=supply].parameters[id=voltage].value	The data sheet gives no minimum VBAT: preconditioning applies below 66.5 % of VREG with no floor, so the pin follows the cell from 0 V up to VREG (4.232 V max). · https://ww1.microchip.com/downloads/aemDocuments/documents/APID/ProductDocuments/DataSheets/MCP73
A Assumption	5.0 V	def:st-usblc6-2sc6:interfaces[id=pin_5].parameters[id=voltage].value	The datasheet states no recommended operating voltage. Leakage is specified at VRM = 5.25 V, which the Rev 4 change note ties to the USB standard, and VBR is 6 V min; 5.25 V is therefore the highest VBUS the stated figures cover. · https://static.chipdip.ru/li
A Assumption	5.0 V	def:st-usblc6-2sc6:interfaces[id=vbus_clamp].parameters[id=voltage].value	The datasheet states no recommended operating voltage. Leakage is specified at VRM = 5.25 V, which the Rev 4 change note ties to the USB standard, and VBR is 6 V min; 5.25 V is therefore the highest VBUS the stated figures cover. · https://static.chipdip.ru/li
D Derived	0	derived:diagCount(nominal,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(nominal,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	7	derived:diagCount(nominal,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	2	derived:diagCount(hr-sensor-on-3v0,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(hr-sensor-on-3v0,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	6	derived:diagCount(hr-sensor-on-3v0,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	3	derived:diagCount(display-on-vbat,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(display-on-vbat,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	7	derived:diagCount(display-on-vbat,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(no-i2c-pullups,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	2	derived:diagCount(no-i2c-pullups,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	7	derived:diagCount(no-i2c-pullups,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(cell-leads-reversed,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(cell-leads-reversed,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	7	derived:diagCount(cell-leads-reversed,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
D Derived	5	derived:diagCount(ldo-output-to-vbat,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	0	derived:diagCount(ldo-output-to-vbat,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	7	derived:diagCount(ldo-output-to-vbat,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(stat-overdrive,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(stat-overdrive,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	7	derived:diagCount(stat-overdrive,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
– Not in model	—	sys:bom[line=1].verification	UHD bom
– Not in model	—	sys:bom[line=2].verification	UHD bom
– Not in model	—	sys:bom[line=6].manufacturer	UHD bom
– Not in model	—	sys:bom[line=6].verification	UHD bom
– Not in model	—	sys:bom[line=7].manufacturer	UHD bom
– Not in model	—	sys:bom[line=7].verification	UHD bom
– Not in model	—	sys:bom[line=8].manufacturer	UHD bom
– Not in model	—	sys:bom[line=8].verification	UHD bom
– Not in model	—	sys:bom[line=14].manufacturer	UHD bom
– Not in model	—	sys:bom[line=14].verification	UHD bom
– Not in model	—	sys:bom[line=26].verification	UHD bom
– Not in model	—	sys:bom[line=29].verification	UHD bom
– Not in model	—	sys:bom[line=30].manufacturer	UHD bom
– Not in model	—	sys:bom[line=30].verification	UHD bom
– Not in model	—	sys:bom[line=38].verification	UHD bom
– Not in model	—	sys:bom[line=39].verification	UHD bom
– Not in model	—	sys:bom[line=40].manufacturer	UHD bom
– Not in model	—	sys:bom[line=40].verification	UHD bom
– Not in model	—	sys:bom[line=41].manufacturer	UHD bom
– Not in model	—	sys:bom[line=41].verification	UHD bom
– Not in model	—	sys:bom[line=42].verification	UHD bom
– Not in model	—	sys:bom[line=43].manufacturer	UHD bom

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
– Not in model	—	sys:bom[line=43].verification	UHD bom
– Not in model	—	sys:bom[line=44].verification	UHD bom
– Not in model	—	sys:bom[line=45].manufacturer	UHD bom
– Not in model	—	sys:bom[line=45].verification	UHD bom